

CLIMATE CHANGE SCIENCE GREENHOUSE EFFECT

GREENHOUSE-EFFECT FIREWALL → ENERGY-BALANCE RAIL → EMISSION-TO-CONCENTRATION LEDGER → FORCING-FEEDBACK FORK → WATER AND PARTICLE MATRIX → GAS COMPARISON GATE

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

PRIMARY CORE FLOW SUBORDINATE EXTRA APPROVAL / REVIEW

Approval: FALSE • Pending user review • active generation g1 • explicit approval of this exact generation is required • all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

00

STAGE 00 GREENHOUSE-EFFECT FIREWALL

GREENHOUSE-EFFECT FIREWALL NATURAL EFFECT ABSORBS AND RE-EMITS OUTGOING LONGWAVE RADIATION MAINTAINS A HABITABLE ENERGY BALANCE HUMAN ENHANCEMENT RAISES CONCENTRATIONS AND NET FORCING CLIMATE RESPONSE WARMING PLUS WIDER SYSTEM CHANGES

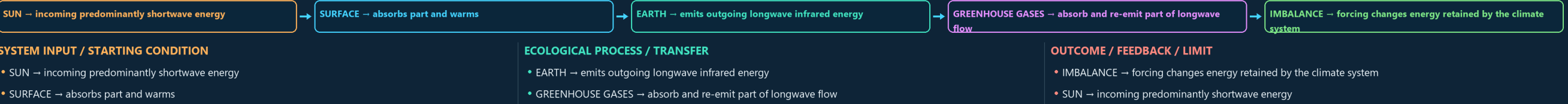
ECOLOGICAL UNIT / PARAMETER	DECISIVE DISTINCTION	SCALE / STATUS / EXCEPTION
NATURAL EFFECT → absorbs and re-emits outgoing longwave radiation	HUMAN ENHANCEMENT → raises concentrations and net forcing	EXAM RULE → preserve the natural-versus-enhanced distinction
FUNCTION → maintains a habitable energy balance	CLIMATE RESPONSE → warming plus wider system changes	NATURAL EFFECT → absorbs and re-emits outgoing longwave radiation
ECOLOGICAL UNIT / PARAMETER - NATURAL EFFECT → absorbs and re-emits outgoing longwave radiation - FUNCTION → maintains a habitable energy balance	DECISIVE DISTINCTION - HUMAN ENHANCEMENT → raises concentrations and net forcing - CLIMATE RESPONSE → warming plus wider system changes	SCALE / STATUS / EXCEPTION - EXAM RULE → preserve the natural-versus-enhanced distinction - NATURAL EFFECT → absorbs and re-emits outgoing longwave radiation

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CLIMATE RESPONSE → warming plus wider system changes.

01

STAGE 01 ENERGY-BALANCE RAIL

ENERGY-BALANCE RAIL INCOMING PREDOMINANTLY SHORTWAVE ENERGY ABSORBS PART AND WARMS EMITS OUTGOING LONGWAVE INFRARED ENERGY GREENHOUSE GASES ABSORB AND RE-EMIT PART OF LONGWAVE FLOW FORCING CHANGES ENERGY RETAINED BY THE CLIMATE SYSTEM

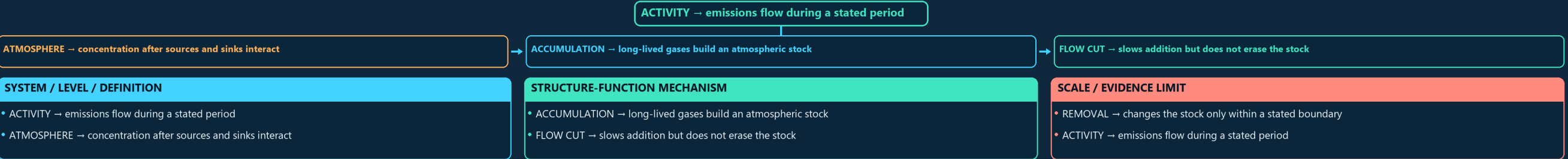


ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: IMBALANCE → forcing changes energy retained by the climate system.

02

STAGE 02 EMISSION-TO-CONCENTRATION LEDGER

EMISSION-TO-CONCENTRATION LEDGER EMISSIONS FLOW DURING A STATED PERIOD CONCENTRATION AFTER SOURCES AND SINKS INTERACT LONG-LIVED GASES BUILD AN ATMOSPHERIC STOCK FLOW CUT SLOWS ADDITION BUT DOES NOT ERASE THE STOCK CHANGES THE STOCK ONLY WITHIN A STATED BOUNDARY

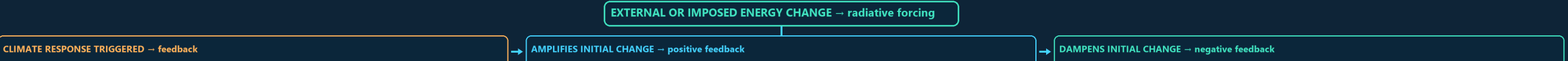


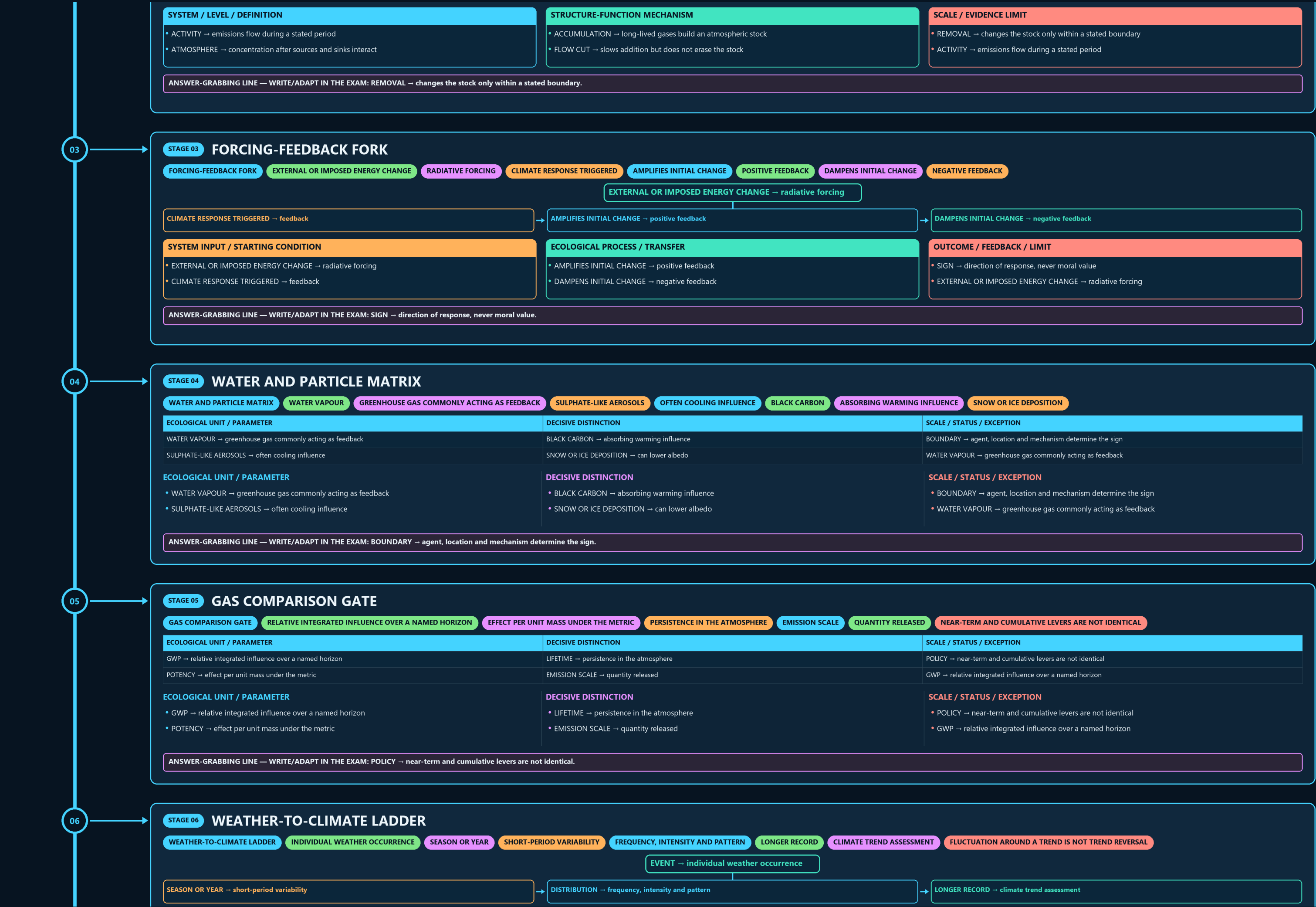
ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: REMOVAL → changes the stock only within a stated boundary.

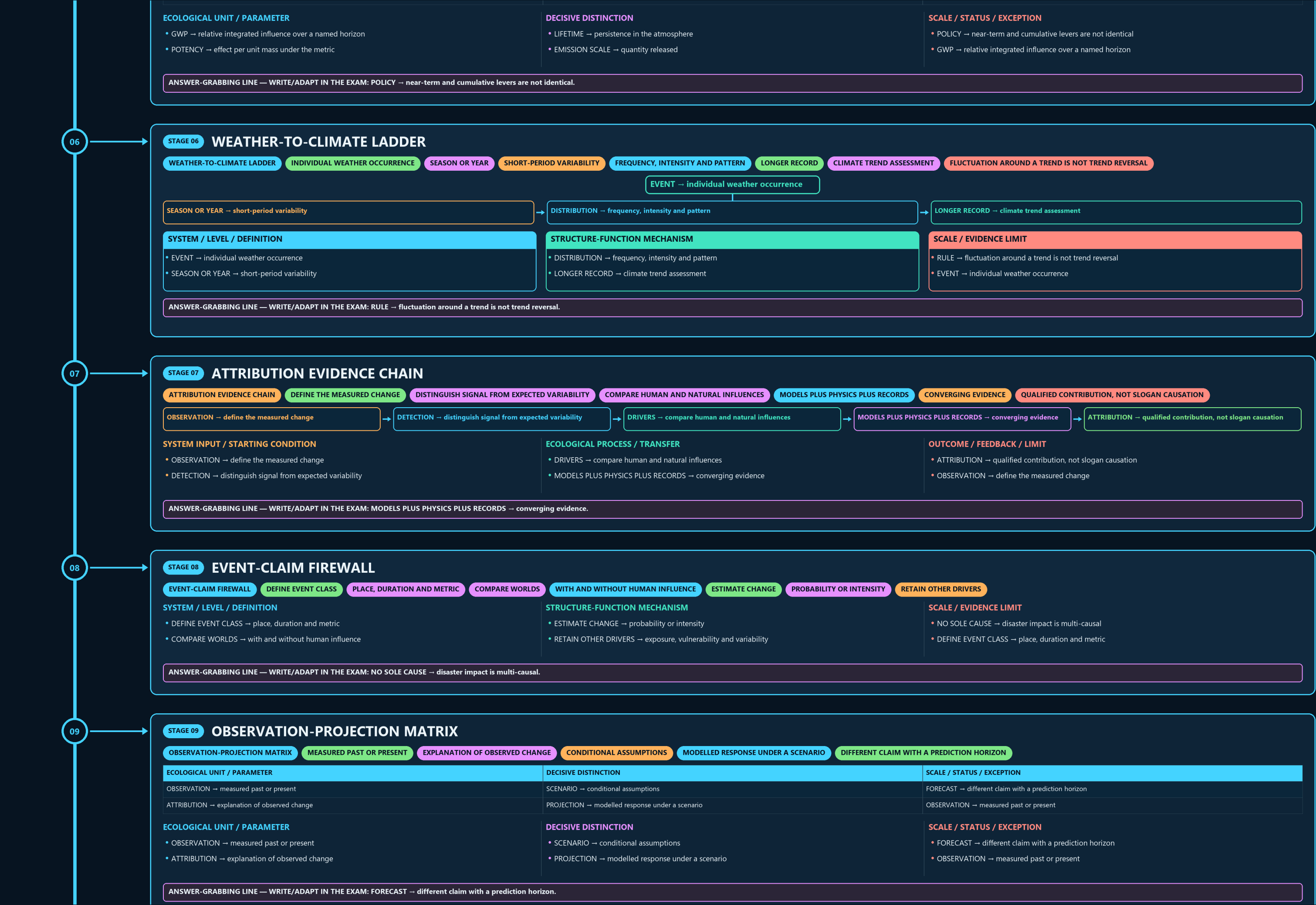
03

STAGE 03 FORCING-FEEDBACK FORK

FORCING-FEEDBACK FORK EXTERNAL OR IMPOSED ENERGY CHANGE RADIATIVE FORCING CLIMATE RESPONSE TRIGGERED AMPLIFIES INITIAL CHANGE POSITIVE FEEDBACK DAMPENS INITIAL CHANGE NEGATIVE FEEDBACK







09

STAGE 09 OBSERVATION-PROJECTION MATRIX

OBSERVATION-PROJECTION MATRIX

MEASURED PAST OR PRESENT

EXPLANATION OF OBSERVED CHANGE

CONDITIONAL ASSUMPTIONS

MODELLED RESPONSE UNDER A SCENARIO

DIFFERENT CLAIM WITH A PREDICTION HORIZON

ECOLOGICAL UNIT / PARAMETER

OBSERVATION → measured past or present

ATTRIBUTION → explanation of observed change

DECISIVE DISTINCTION

SCENARIO → conditional assumptions

PROJECTION → modelled response under a scenario

SCALE / STATUS / EXCEPTION

FORECAST → different claim with a prediction horizon

OBSERVATION → measured past or present

ECOLOGICAL UNIT / PARAMETER

• OBSERVATION → measured past or present

• ATTRIBUTION → explanation of observed change

DECISIVE DISTINCTION

• SCENARIO → conditional assumptions

• PROJECTION → modelled response under a scenario

SCALE / STATUS / EXCEPTION

• FORECAST → different claim with a prediction horizon

• OBSERVATION → measured past or present

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: FORECAST → different claim with a prediction horizon.

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STAGE 10 CLIMATE-RESPONSE FORK

CLIMATE-RESPONSE FORK

SOURCE REDUCTION

SINK ENHANCEMENT

MITIGATION WITH ACCOUNTING BOUNDARY

EXPOSURE OR VULNERABILITY REDUCTION

RESIDUAL HARM

SEPARATE LOSS-AND-DAMAGE QUESTION

MITIGATION AND ADAPTATION ARE COMPLEMENTARY

SOURCE REDUCTION → mitigation

SINK ENHANCEMENT → mitigation with accounting boundary

EXPOSURE OR VULNERABILITY REDUCTION → adaptation

RESIDUAL HARM → separate loss-and-damage question

SYSTEM / LEVEL / DEFINITION

• SOURCE REDUCTION → mitigation

• SINK ENHANCEMENT → mitigation with accounting boundary

STRUCTURE-FUNCTION MECHANISM

• EXPOSURE OR VULNERABILITY REDUCTION → adaptation

• RESIDUAL HARM → separate loss-and-damage question

SCALE / EVIDENCE LIMIT

• PORTFOLIO → mitigation and adaptation are complementary

• SOURCE REDUCTION → mitigation

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RESIDUAL HARM → separate loss-and-damage question; PORTFOLIO → mitigation and adaptation are complementary.

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STAGE 11 SCIENCE ANSWER SPINE

SCIENCE ANSWER SPINE

NATURAL EFFECT, HUMAN ENHANCEMENT AND ENERGY FLOW

EMISSIONS, CONCENTRATIONS, STOCK AND NET BALANCE

FORCING, FEEDBACK AND GAS-SPECIFIC PROPERTIES

TREND, DETECTION, ATTRIBUTION AND PROJECTION

SCALE, SCENARIO, CONFIDENCE AND NO INVENTED FIGURES

DEFINE → natural effect, human enhancement and energy flow

ACCOUNT → emissions, concentrations, stock and net balance

EXPLAIN → forcing, feedback and gas-specific properties

EVALUATE → trend, detection, attribution and projection

QUALIFY → scale, scenario, confidence and no invented figures

SYSTEM / LEVEL / DEFINITION

• DEFINE → natural effect, human enhancement and energy flow

• ACCOUNT → emissions, concentrations, stock and net balance

STRUCTURE-FUNCTION MECHANISM

• EXPLAIN → forcing, feedback and gas-specific properties

• EVALUATE → trend, detection, attribution and projection

SCALE / EVIDENCE LIMIT

• QUALIFY → scale, scenario, confidence and no invented figures

• DEFINE → natural effect, human enhancement and energy flow

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: QUALIFY → scale, scenario, confidence and no invented figures.

E

EXTRA SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT

INDIA'S NATIONAL EMISSIONS PROFILE (AS REPORTED IN ITS PERIODIC

ANALYTICAL POINT

INDIA'S PER-CAPITA AND HISTORICAL CUMULATIVE EMISSIONS REMAIN

GLOBAL EMISSIONS AND REMAINING-CARBON-BUDGET FIGURES ARE PERIODICALLY UPDATED BY

ATTRIBUTION-SCIENCE CONFIDENCE LEVELS VARY MEANINGFULLY BY EVENT TYPE (HIGH

INDIA-SPECIFIC SECTORAL EMISSIONS DATA (FROM NATIONAL COMMUNICATIONS/BURS) IS REPORTED

IPCC AR6 ATTRIBUTION SCIENCE HAS SUBSTANTIALLY STRENGTHENED CONFIDENCE IN

OPTIONAL ECOLOGICAL PROCESS DEPTH

• CAUTION: India's national emissions profile (as reported in its periodic UNFCCC national

• CAUTION: Analytical point: India's per-capita and historical cumulative emissions remain

• CAUTION: Global emissions and remaining-carbon-budget figures are periodically updated by IPCC

• CAUTION: Attribution-science confidence levels vary meaningfully by event type (high confidence

SCALE, PARAMETER OR STATUS LIMIT

• CAUTION: India-specific sectoral emissions data (from national communications/BURS) is reported

• IPCC AR6 attribution science has substantially strengthened confidence in linking

• Aerosols exert a net cooling (negative radiative forcing) effect that partially masks

• The remaining carbon budget concept reframes climate policy around a finite, depleting

COMPARATIVE RESTORATION NUANCE

• India's methane and nitrous oxide emissions are disproportionately driven by

• NOT: All extreme weather events can be attributed to climate change with equally high

• NOT: Reducing air pollution (aerosols) has no bearing on the rate of global warming. - It

• NOT: A distant net-zero target date alone is sufficient to remain within a fixed carbon

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.